

Aligning Large Language Models for Psychiatric Safety Using Expert-Annotated Constraints

Abstract

This study outlines the development of a highly specialized conversational AI designed to support patients with eating disorders and perioperative mental health challenges. By adapting the Articulate Medical Intelligence Explorer (AMIE) architecture, this research addresses the critical need for clinical safety in mental health chatbots. Utilizing a unique dataset of clinician-patient transcripts annotated with multi-dimensional expert tags for suboptimal responses, the study proposes a novel dual-stage training pipeline. This involves data augmentation via persona-driven simulated role-play, followed by parallel experiments utilizing three distinct alignment methods: Attribute-Conditioned Supervised Fine-Tuning (SFT), Kahneman-Tversky Optimization (KTO), and Direct Preference Optimization (DPO). Furthermore, the project optimizes the underlying AMIE chain-of-reasoning by introducing a Structured Chain-of-Thought (SCoT) to minimize latency and reduce hallucinations. The resulting architecture aims to deliver a chatbot that is mathematically constrained for clinical safety while remaining conversationally fluid.

1. Introduction and Background

The integration of Large Language Models (LLMs) into healthcare has catalyzed a paradigm shift in how automated diagnostic and support systems are designed. Recent milestones, such as the Articulate Medical Intelligence Explorer (AMIE), have demonstrated that LLMs equipped with chain-of-reasoning capabilities can conduct highly articulate and accurate diagnostic dialogues. However, the expansion of these technologies into the psychiatric domain—specifically addressing eating disorders and perioperative mental health—introduces profound complexities. Mental health interventions require not only factual accuracy but also high degrees of emotional intelligence, nuance, and strict adherence to clinical safety protocols, as a single insensitive response can cause measurable psychological harm.

The background to this problem is rooted in the distinct differences between physiological medical diagnosis and psychiatric conversational support. The original AMIE architecture relies heavily on Supervised Fine-Tuning (SFT) over uncensored transcripts and simulated patient role-play to develop its conversational capabilities. In general medicine, extracting symptoms and matching them to a pathology is a relatively objective task. In contrast, mental health chatbots must navigate highly subjective human experiences. A chatbot assisting a patient with an eating disorder must actively avoid multi-dimensional conversational pitfalls, such as toxic positivity, emotional invalidation, or offering prescriptive dietary advice that violates psychiatric guidelines. Standard LLMs inherently lack this specialized clinical intuition and,

without explicit boundaries, are prone to generating text that is superficially polite but clinically harmful.

The primary aim of this research is to adapt the AMIE architecture for the mental health domain by leveraging a dataset of transcripts annotated with highly specific, multi-dimensional negative tags. By substituting standard alignment methods with techniques explicitly designed for unpaired data, granular constraints, or synthesized pairs, this study seeks to mathematically map clinical boundaries into the model’s policy, bridging the gap between general medical AI and specialized psychiatric support systems.

2. Problem Statement and Motivation

The central research problem is that prevailing methodologies for training diagnostic LLMs rely on the implicit assumption that the training corpus represents universally optimal behavior, rendering them dangerously inadequate for mental health applications. Our dataset uniquely highlights this issue: it contains explicitly tagged suboptimal clinician responses, demonstrating that even human-generated medical dialogues contain critical errors.

Furthermore, these errors are *multi-dimensional*. A response may be flagged for ‘toxic positivity’ but not ‘prescriptive advice.’ Traditional alignment algorithms typically flatten this rich nuance into a single “bad” label and rely on paired “correct” counterparts that do not naturally exist in our raw dataset. Therefore, there is an imperative need to develop a methodology that can successfully align a reasoning model by natively leveraging unpaired, multi-tagged negative data, or by intelligently synthesizing missing correct pairings to facilitate direct optimization.

Deploying standard conversational models in psychiatry is highly irresponsible due to their propensity for hallucination and emotional invalidation. By engineering an architecture that explicitly understands and avoids the distinct conversational triggers associated with mental distress, healthcare providers can safely deploy AI triage and support tools that genuinely protect patient well-being while addressing critical shortages in human clinical availability.

3. Research Questions and Objectives

To address the challenges of adapting medical reasoning models for psychiatric use, this study poses the following research questions:

1. How can the AMIE chain-of-reasoning architecture be modified using a Structured Chain-of-Thought (SCoT) approach to ensure sub-second latency and factual grounding in real-time psychiatric dialogues?
2. Which alignment method—Attribute-Conditioned SFT, Kahneman-Tversky Optimization (KTO), or Direct Preference Optimization (DPO)—most effectively leverages explicitly tagged negative data to suppress harmful conversational outputs?
3. To what extent does a system-level multi-agent architecture (incorporating input/output guardrails) improve the holistic clinical safety of the model compared to relying solely on weight-based alignment?

Directly aligned with these questions, the objectives of this research are:

1. To adapt the AMIE inner-monologue framework into a Structured Chain-of-Thought (SCoT) to reduce real-time inference latency using high-throughput engines like vLLM.
2. To evaluate and compare the efficacy of advanced unpaired and multi-attribute alignment techniques against direct preference frameworks.
3. To quantify the reduction in critical clinical dialogue failures through automated semantic metrics evaluated on a held-out test set of our expert-annotated corpus, validated further by blinded human-in-the-loop clinical evaluations.

4. Literature Review and Theoretical Framework

The foundational benchmark for medical conversational AI is established by Tu et al. (2024) with the AMIE model, which utilizes simulated environments and free-form chain-of-thought reasoning (Wei et al., 2022) to conduct diagnostic dialogue. However, AMIE’s reliance on unstructured reasoning and standard Supervised Fine-Tuning (SFT) presents significant latency and clinical safety risks when applied to highly sensitive psychiatric settings. To mitigate this, recent architectural advancements point toward Structured Chain-of-Thought (SCoT), which constrains the reasoning process into predictable, machine-readable formats (e.g., JSON). This structural constraint is vital for psychiatric applications, as it drastically reduces inference latency and enables deterministic, programmatic safety checks prior to generating a natural language response.

Beyond structural reasoning modifications, aligning a medical model to navigate multi-dimensional psychological boundaries requires specific architectural choices. Consequently, this study explores three advanced alignment paradigms.

First, **Attribute-Conditioned SFT** builds on methodologies like SteerLM (Dong et al., 2023), proving that models can be effectively aligned by embedding explicit multi-dimensional attributes directly as control tokens during SFT, granting granular steerability away from negative traits during inference. Second, **Kahneman-Tversky Optimization (KTO)** (Ethayarajh et al., 2024) sidesteps the need for paired data entirely. Grounded in Prospect Theory, KTO maps distinct negative tags into an aggregate negative utility signal, optimizing the model’s policy directly from unpaired data by heavily penalizing suboptimal responses.

For the third paradigm, standard optimization frameworks must be adapted to function on un-paired data via intelligent data synthesis. **Direct Preference Optimization (DPO)** acts as a highly efficient alignment engine. It utilizes oracle-generated preference pairs to optimize the policy directly (Rafailov et al., 2023), bypassing the need for complex reinforcement learning loops entirely.

5. Research Methodology

The methodology consists of a three-phase computational approach designed to adapt the foundational AMIE architecture securely and efficiently for multi-tag psychiatric data.

Phase 1: Baseline SFT and Iterative Role-Play Augmentation. To adapt the base model (e.g., Llama-3) for the specialized psychiatric domain, this phase utilizes an iterative fine-tuning loop adapted from the AMIE methodology.

First, the base model will undergo an initial round of Supervised Fine-Tuning (SFT) on our static, high-quality data mixture. This includes the expert-annotated clinician transcripts and open-source emotional support datasets (e.g., DAIC-WOZ, CounselChat, ESConv). Crucially, during this initial SFT, we will retroactively inject the Structured Chain-of-Thought (SCoT) formatting into the training targets. This establishes a baseline model capable of empathetic dialogue and structural JSON state-tracking.

Once the baseline Clinician Agent is established, we will initiate multiple rounds of **Persona-Driven Simulated Role-Play** to dynamically scale the training corpus. We will employ an advanced oracle model (e.g., GPT-4) to act as both a Patient Agent and a Critic Agent. The system will programmatically generate synthetic patient personas with specific psychiatric vulnerabilities (e.g., eating disorder denial, perioperative panic), instructing the Patient Agent to inject conversational stress-tests and demand unsafe advice.

During each round of data generation, the baseline Clinician Agent and Patient Agent complete a full dialogue. The Critic Agent then evaluates the transcript against a strict clinical safety rubric, identifying multi-dimensional errors (e.g., emotional invalidation). The Clinician Agent then restarts the dialogue with the same Patient Agent *from scratch*. During this new attempt, the Clinician Agent retains in-context access to both the previous failed dialogue history and the Critic’s explicit feedback, allowing it to dynamically course-correct. This self-improvement process is repeated twice per scenario to generate a highly polished, clinically safe transcript.

Finally, the intermediate rough drafts and Critic prompts are discarded, and the Clinician model undergoes another iteration of SFT strictly on these newly perfected transcripts. By repeating this entire cycle (Role-Play → Critic Feedback → Restart and Refine → SFT) across multiple rounds, the model progressively bootstraps its own clinical reasoning and psychiatric safety guardrails.

Phase 2: Advanced Alignment Comparisons. To enforce safety constraints using our multi-tagged negative data, we will conduct three parallel alignment experiments. The final method requires a preliminary data synthesis step to create preference pairs from our unpaired negatives by prompting an oracle LLM to safely rewrite the erroneous responses.

- **Attribute-Conditioned SFT:** We will embed the specific multi-dimensional expert tags as system-level control tokens during training. At inference, the prompt will be explicitly set to demand zero negative traits, steering the model away from harmful generations.
- **Kahneman-Tversky Optimization (KTO):** We will aggregate the nuanced multi-tags into a composite negative utility signal. By mapping these sub-optimal responses to a severe penalty, we will utilize KTO to optimize the policy directly from unpaired data, testing if Prospect Theory’s aggregate mathematical elegance outperforms multi-tag conditioning.
- **Direct Preference Optimization (DPO):** Utilizing oracle-synthesized preference pairs,

we will execute standard DPO. This provides a direct contrast to unpaired methods like KTO, testing whether synthesizing explicit “correct” and “incorrect” response pairs for direct policy optimization can maintain strict clinical safety boundaries without sacrificing conversational fluidity.

Phase 3: Architectural Robustness and Low-Latency Inference. To address the latency inherent in AMIE’s free-form chain-of-reasoning, the model will employ a **Structured Chain-of-Thought (SCoT)**. This approach bridges machine-readable and human-readable reasoning: before responding, the model is forced to generate a hidden, strictly constrained JSON object tracking immediate clinical risk flags, followed by a heavily token-limited natural language rationale (e.g., max 20 tokens). This combination enables programmatic safety checks while maintaining clinical nuance. As a potential secondary goal for this phase, we may also explore integrating a lightweight Retrieval-Augmented Generation (RAG) module to fetch up-to-date psychiatric protocols during the reasoning step. Inference will be optimized using vLLM and speculative decoding. The efficacy of this architecture and the comparative success of the alignment paradigms will be measured using the comprehensive framework outlined in the following section.

6. Evaluation Metrics

To rigorously quantify the efficacy of the three alignment strategies and the structural architectural changes, evaluation will be split across three distinct modalities:

- **Automated Semantic and Safety Metrics:** We will evaluate the model’s responses to complex clinical dialogues sampled from a strictly held-out test set of our unique expert-annotated dataset. We will employ an LLM-as-a-Judge framework (utilizing an advanced oracle model, such as GPT-4) prompted with a strict clinical rubric. The judge will score generated responses on a scale of 1 to 5 for various metrics, empathy, safety, and therapeutic goal-adherence, explicitly testing for the successful suppression of the multi-dimensional negative tags.
- **Human-in-the-Loop Clinical Evaluation:** A panel of blinded mental health professionals will conduct A/B testing on the generated responses against the base model and baseline AMIE behaviors.
- **System and Latency Profiling:** To ensure the SCoT modifications do not impede real-time conversational flow, Time-To-First-Token (TTFT) and overall inference latency will be mathematically profiled across the vLLM engine. The system will target a strict sub-second response threshold suitable for synchronous, low-friction patient support.

References

- [1] Dong, Y., et al. (2023). SteerLM: Attribute Conditioned SFT as an (User-Steerable) Alternative to RLHF. *arXiv preprint arXiv:2310.05344*.
- [2] Ethayarajh, K., Xu, W., Jurafsky, D., & Kiela, D. (2024). KTO: Model Alignment as Prospect Theoretic Optimization. *arXiv preprint arXiv:2402.01306*.
- [3] Rafailov, R., Sharma, A., Mitchell, E., et al. (2023). Direct preference optimization: Your language model is secretly a reward model. *Advances in Neural Information Processing Systems*, 36.
- [4] Tu, T., Palepu, A., Schaekermann, M., et al. (2024). Towards Conversational Diagnostic AI. *arXiv preprint arXiv:2401.05654*.
- [5] Wei, J., Wang, X., Schuurmans, D., et al. (2022). Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35, 24824-24837.